

Yuanzhu Zhan

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🌐 Yuanzhu Zhan | 📄 YuanzhuZhan | 🎓 Google Scholar

EDUCATION

Pennsylvania State University

PhD in Aerospace Engineering

Aug. 2024 – Present

State College, U.S.

- GPA: 3.97/4.00

Nankai University

Bachelor of Engineering in Intelligence Science and Technology

Sep. 2020 – Jun. 2024

Tianjin, China

- GPA: 3.78/4.00

RESEARCH INTERESTS

- **Aerial autonomy:** Closing the perception, planning, and control loop for aerial robots to enable robust autonomy in unstructured environments.
- **Dexterous, contact-rich aerial manipulation:** Enabling aerial robots to interact precisely and reliably with objects and surfaces through contact-aware perception and control.

PUBLICATIONS

- [C.1] Yuanzhu Zhan*, Yufei Jiang*, Zemu Zhang, Junyi Geng. (2026). **Aerial Manipulation in the Wild with Onboard Perception, Policy Learning, and Whole-Body Control**. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 5th Workshop on Mobile Manipulation and Embodied Intelligence (MoMA.v5)*. Accepted.
- [C.2] Yutong Wang, Dongjae Lee, Xiaofeng Guo, **Yuanzhu Zhan**, Yufei Jiang, Bavin Saravanan, Muqing Cao, Jia Xie, Chenyang Mao, Sebastian Scherer, Junyi Geng, Guanya Shi. (2026). **AM-Bench: A Modular Simulation Suite and Benchmark for Aerial Manipulation Policy Learning**. *Conference on Robot Learning (CoRL)*. Accepted.
- [C.3] Yufei Jiang*, **Yuanzhu Zhan***, Harsh Vardhan Gupta, Chinmay Borde, Junyi Geng. (2026). **A Self-Supervised Learning Approach with Differentiable Optimization for UAV Trajectory Planning**. *IEEE International Conference on Robotics and Automation (ICRA)*. Accepted.
- [C.4] **Yuanzhu Zhan***, Yufei Jiang*, Muqing Cao, Junyi Geng. (2026). **Aerial Manipulation with Contact-Aware Onboard Perception and Hybrid Control**. *IEEE International Conference on Robotics and Automation (ICRA)*. Accepted.
- [C.5] Jinyu Liu, **Yuanzhu Zhan**, Wenduo Jia, Jianxun Zhang, Yu Dai. (2025). **A Surgical State Identifying Method based on BiLSTM with Vibration Processing for Improving Safety of Bone Milling System**. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 4018–4023.

ACADEMIC SERVICE

- **Conference Reviewer:** ICRA 2026
- **Journal Reviewer:** IEEE Robotics and Automation Letters (RA-L) 2025, 2026
- **Journal Reviewer:** IEEE Transactions on Robotics (T-RO) 2026
- **Journal Reviewer:** Journal of Guidance, Control, and Dynamics 2025

SKILLS

- **Programming and software:** C++, Python, MATLAB, HTML, JavaScript, PyTorch, Gazebo, Isaac Lab, ROS/ROS2, SolidWorks, Fusion 360, Adobe PS/PR/AI

- **Hardware:** NVIDIA Jetson, Pixhawk, ModalAI VOXL, RealSense/OAK/ZED cameras
- **Experimental skills:** 3D printing, drilling, soldering
- **Languages:** Mandarin (native), English (fluent)